



TAMPINES JUNIOR COLLEGE

Preliminary Examinations 2008

CHEMISTRY (Higher 2)

9746/2

PAPER 2 Structured

Thursday

21 August 2008

1 hour 30 minutes

Additional materials: Data Booklet

INSTRUCTIONS TO CANDIDATES

Write your answers on the Question Paper.

Write your name and Civics Group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The number of marks is given in brackets [] at the end of each question or part question.

Name : _____

CG : _____

Question	Marks
1	
2	
3	
4	
5	
TOTAL	/ 60

This document consists of 7 printed pages.

[Turn over

Answer **all** questions.

1 (a) 2.09 g of K_2SO_4 was dissolved in 50 cm^3 of water. 4.98 g of BaCl_2 was then added.

(i) What would be observed upon addition of the barium chloride?

_____ [1]

(ii) Explain your answer to (a)(i) given that the solubility product of BaSO_4 is $1.08 \times 10^{-10} \text{ mol}^2 \text{ dm}^{-6}$.

[3]

(b) The melting point of BaSO_4 was found to be 1580°C . Magnesium sulphate, MgSO_4 , however, decomposes at 1124°C , even before melting, to give a sulphur oxide. Suggest why MgSO_4 decomposes more readily.

_____ [2]

(c) Draw a dot and cross diagram to represent the bonding in BaSO_4 .

[2]

[Total: 8]

2 Aluminium is obtained by electrolysis of molten aluminium oxide using inert electrodes.

(a) Give ion-electron equations for the electrode processes:

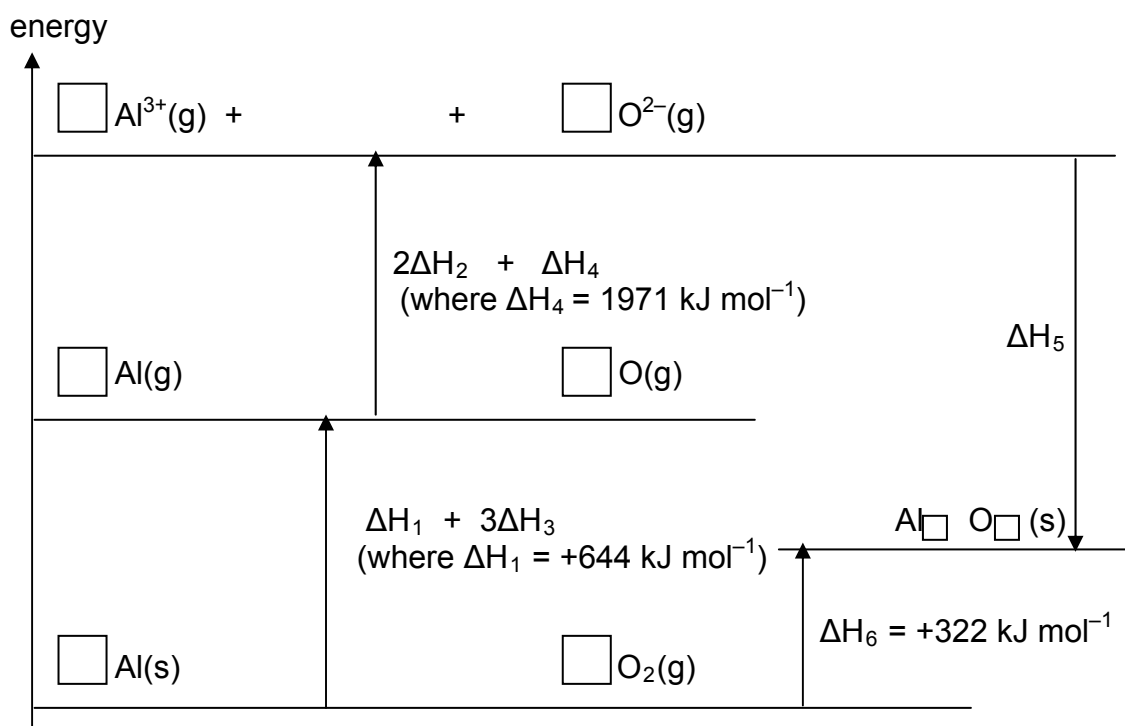
at the cathode: _____

at the anode: _____ [2]

(b) Calculate the mass of aluminium produced and the volume of gas liberated at s.t.p. when a current of 8 A is passed for 100 minutes through the molten aluminium oxide.

[3]

(c) The formation of aluminium oxide from its elements may be represented by a Born-Haber energy cycle as described below.



(i) Fill in the boxes provided with the relevant numbers.

(ii) Use of Data Booklet is required in this question.

Name and calculate the value of the enthalpy changes represented as:
(Note that ΔH_2 is relevant to aluminium while ΔH_3 is relevant to oxygen.)

ΔH_2 _____

ΔH_3 _____

(iii) Hence, name the enthalpy change represented as ΔH_5 and calculate its value.

ΔH_5 : _____

[9]

[Total: 14]

- 3 Calcium ion, Ca^{2+} , is required for blood clotting and other cell processes. An abnormal Ca^{2+} concentration in the body is indicative of disease.
To measure Ca^{2+} concentration, 1 cm^3 of human blood was treated with sodium ethanedioate ($\text{Na}_2\text{C}_2\text{O}_4$) solution. The resulting CaC_2O_4 precipitate was filtered and dissolved in dilute acid. This solution then required 20.50 cm^3 of $4.88 \times 10^{-5} \text{ mol dm}^{-3}$ KMnO_4 for complete reaction.

(a) (i) The dark purple colour of the potassium manganate(VII), makes it suitable for its use as described above.

Explain why transition metal compounds are usually coloured.

(ii) State the formula of the manganese-containing species that would be formed in the reaction described above, and hence suggest the colour change that would be observed.

[4]

(b) Carbon dioxide is formed in the reaction between ethanedioate and manganate(VII) ions. Use your answer to part (a)(ii), together with the *Data Booklet* to write an equation for the reaction.

[2]

(c) Determine the amount of Ca^{2+} ions present in the blood sample, and hence, the concentration of Ca^{2+} in terms of mg per 100 ml of blood. [$1 \text{ ml} = 1 \text{ cm}^3$]

[3]

(d) Sodium ethanedioate is the salt formed from the neutralisation of ethanedioic acid by sodium hydroxide.

(i) State the type of hybridisation present in the carbon atoms of the $\text{C}_2\text{O}_4^{2-}$ ion and sketch the hybrid orbitals around the atom.

(ii) Hence state the O–C–O bond angle in the ethanedioate ion.

[3]

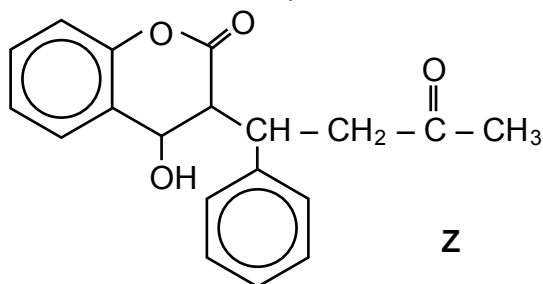
[Total: 12]

4 Describe how chlorine water reacts with each of the following reactants. In each case, state the expected observations, identify the type of reaction occurring and write relevant equations. [12]

Reactant	Observations	Type of reaction	Equations
(a) hot aqueous NaOH			
(b) aqueous KI			
(c) propane, in the presence of light			
(d) phenylamine			

[Total: 12]

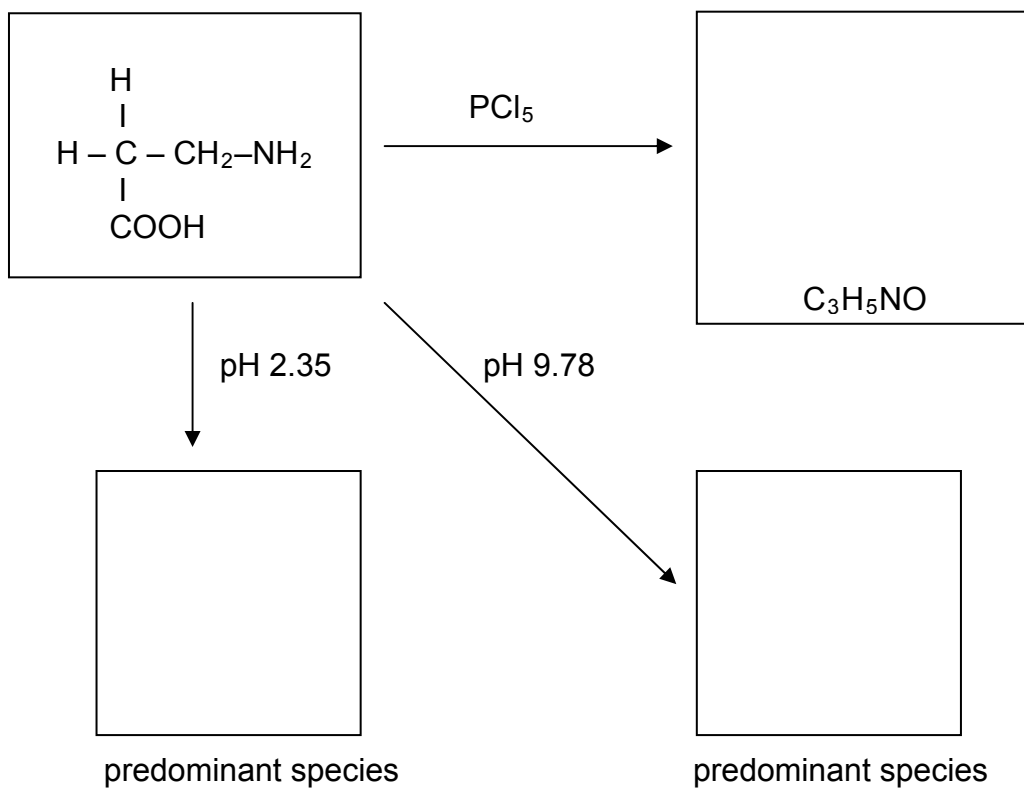
5 The structural formula of a compound **Z** is shown below.



- (a) On the formula above, use an arrow to indicate the bond that will cleave when **Z** undergoes hydrolysis. [1]
- (b) For each of the following reagents, identify the group in molecule **Z** that will react and draw the structural formula(e) of the organic product(s) formed.
(Note: You can use “R” to represent the part of the Z that does not take part in the reaction.) [10]

reagent	group responsible	structural formula(e) of the organic product(s)
alkaline I ₂ (aq)		
H ₂ SO ₄ (aq) under reflux		
2,4 – DNPH		
ethanoyl chloride		
phosphorous pentachloride		

(c) Draw the structural formulae of the organic products formed in the spaces provided. [3]



[Total: 14]

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